

LESSON 5-4 • HOMEWORK PACKET • LEVEL 1 • SHEET 1 OF 2**Area of Regular Polygons**

Maryland Standard 6.G.A.1 • Learning target – I can find the area of a regular polygon by finding the area of one triangle and multiplying.

REMEMBER

- **Regular polygon:** equal sides, equal angles.
- Lines from the center to each vertex make n **congruent triangles**. Each one's height is the **apothem**.
- **$A = n \times (\frac{1}{2} \times \text{side} \times \text{apothem})$.** Use $\approx$ if rounded.

WORKED EXAMPLE

Regular hexagon: side 8 cm, apothem ≈ 6.9 cm.

Step 1 A hexagon makes 6 congruent triangles.

Step 2 One triangle: $\frac{1}{2} \times 8 \times 6.9 = 27.6$

Step 3 Six triangles: $6 \times 27.6 = 165.6$

Answer $\approx 165.6 \text{ cm}^2$

WORD BANK

regular polygon – all sides equal, all angles equal

apothem – center to the middle of a side

congruent – same size and same shape
 $\approx$ – about

PRACTICE

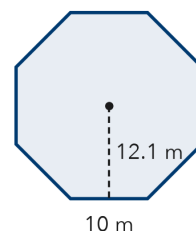
show your work • label units

1. Find the area of the regular octagon. The apothem is rounded.

Step 1 Find one triangle: $\frac{1}{2} \times 10 \times 12.1$.

Step 2 An octagon has 8 triangles. Multiply by 8.

Show your work.



Answer _____ m^2

2. Complete the table. Each apothem is rounded, so mark each area with $\approx$.

Step 1 Perimeter = number of sides $\times$ side.

Step 2 Area $\approx \frac{1}{2} \times \text{perimeter} \times \text{apothem}$.

Regular polygon	Side	Apothem	Perimeter	Area
Hexagon	4 cm	3.5 cm		
Pentagon	10 ft	6.9 ft		

3. Each regular polygon has perimeter P and rounded apothem a . Find each area. Mark it with $\approx$.

a. pentagon: $P = 30 \text{ cm}$, $a \approx 4.1$
cm _____ cm^2

b. hexagon: $P = 60 \text{ ft}$, $a \approx 8.7 \text{ ft}$ _____ ft^2

4. **Gazebo Floor.** A gazebo floor is a regular **hexagon**. Each side is 6 ft. The apothem is about 5.2 ft. Tile costs \$3 for each square foot. About how much does the tile cost?

Step 1 Find the area: $6 \times (\frac{1}{2} \times 6 \times 5.2)$.

Step 2 Multiply the area by \$3.

Show your work.

Answer _____

5. **Find the Error.** Andre says a regular pentagon with sides of 8 cm and an apothem of 5.5 cm has area $\frac{1}{2} \times 8 \times 5.5 = 22 \text{ cm}^2$. Explain his mistake and find the area.

Frame Andre found only ___ triangle. A pentagon has ___ triangles, so the area is ___.

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #____

LESSON 5-4 • HOMEWORK PACKET • LEVEL 1 • SHEET 2 OF 2

Composite Figures

Maryland Standard 6.G.A.1 • Learning target – I can find the area of a composite figure by adding or subtracting the areas of simpler shapes.

REMEMBER

- A **composite figure** is made of simpler shapes.
- **Add** the areas of joined shapes. **Subtract** a piece that is cut out.
- Use the sides you know to find missing sides.

WORKED EXAMPLE

House front: a 10 ft × 6 ft rectangle with a triangular roof on top. The roof's base is 10 ft and it is 4 ft tall.

Step 1 Rectangle: $10 \times 6 = 60$

Step 2 Triangle: $\frac{1}{2} \times 10 \times 4 = 20$

Step 3 The shapes are joined, so add: $60 + 20 = 80$

Answer 80 ft²

WORD BANK

composite figure – a shape made of simpler shapes

add – put areas together

subtract – take away a cut-out area

right angle – a square corner

PRACTICE

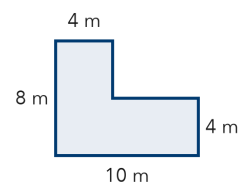
show your work • label units

1. Find the area of the L-shaped figure. All corners are right angles.

Step 1 Cut the L into two rectangles.

Step 2 Find each rectangle's area.

Step 3 Add the two areas.



Show your work.

Answer _____ m²

2. Write **add** or **subtract** to tell how you would find each area.

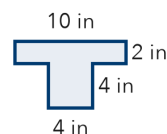
a. a wall, not counting its window: _____ b. a bedroom plus its closet: _____

3. Find the area of the T-shaped figure. All corners are right angles.

Step 1 Cut the T into a top bar and a stem.

Step 2 Find each rectangle's area.

Step 3 Add the two areas.



Answer _____ in²

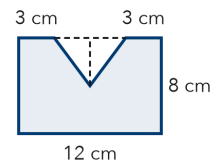
4. A triangle notch 4 cm deep is cut from the top of a rectangle.
Find the area that is left.

Step 1 Find the rectangle: 12×8 .

Step 2 Notch base: $12 - 3 - 3$. Find the triangle.

Step 3 Subtract the triangle.

Answer _____ cm^2



5. **Backyard.** A yard is a **rectangle**, 20 ft by 12 ft. A square garden is in one corner. Each side of the garden is 5 ft. The garden is **not** paved. Find the paved area.

Step 1 Find the yard area.

Step 2 Subtract the garden area.

Answer _____ ft^2

TEST PRACTICE

choose the best answer

6. A rectangle 9 m by 6 m has a square 2 m by 2 m cut out. What area is left?
A 50 m^2 **B** 58 m^2 **C** 54 m^2 **D** 13 m^2
7. A 10 in by 4 in rectangle is joined to a triangle with base 4 in and height 3 in. What is the total area?
A 34 in^2 **B** 40 in^2 **C** 46 in^2 **D** 52 in^2

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____